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SECTION 03

ASSIGNMENT 3:
PARKING DETECTION APPLICATION
THEME:
SMART CITY

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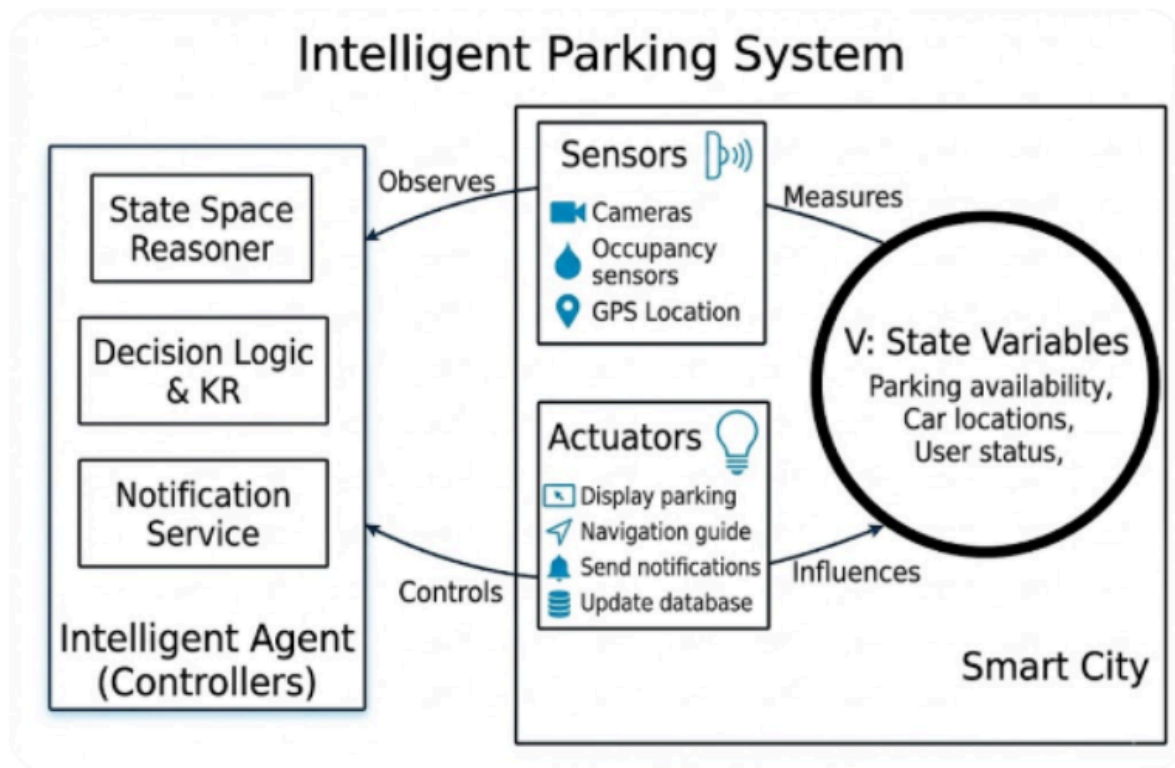
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1.0 PEAS Model

Performance measure	- Accuracy of parking space detection - Speed of real-time database updates - Reduction in parking search time - Timely notifications when parking is full
Environment	- Cities or campuses parking areas - Parking spaces - Vehicles entering and leaving - Driver using the application - Cameras, sensors and network infrastructure
Actuators/Effectors	- Updating the parking database - Displaying available parking spaces on the application - Providing navigation guidance - Sending notifications to drivers when parking is full
Sensors	- Surveillance cameras - Parking occupancy sensors - GPS location data from drivers - User requests from the application - Database status logs

2.0 Relations Between Each Property



Sensors collect real-time data from the parking environment and send it to the intelligent agent. In order to make decisions, the agent uses state-space reasoning and knowledge representation. After that, the actuators carry out tasks like delivering notifications, directing drivers and updating the database. A perception-decision-action loop is created when the environment is constantly changing and giving the sensors feedback.

3.0 Explanation In Proof Of Concept (POC)

Performance:

The performance of the intelligent parking system is measured by how well it supports drivers in finding parking efficiently and accurately. The system aims to provide accurate detection of parking availability, ensuring that occupied and available spaces are correctly identified. Another key performance factor is the speed of the system response, where parking status updates and notifications must be delivered in real time to avoid outdated information. Additionally, the system focuses on reducing parking search time by guiding drivers to available spaces and notifying them promptly when parking areas are full. Timely notifications help drivers make better decisions and prevent unnecessary entry into fully occupied parking areas, improving overall user experience and traffic flow.

Environment:

The environment of the intelligent parking system consists of urban or campus parking areas where vehicles continuously enter and leave. This environment includes parking spaces, vehicles, and drivers who interact with the system through a mobile application. The parking environment is dynamic, as parking availability changes frequently depending on the vehicle movement. The system also operates within a technological environment supported by cameras, parking sensors, GPS signals, network connectivity, and databases. These components allow the intelligent agent to observe changes in parking availability and respond accordingly. Because the environment is constantly changing, continuous monitoring is required to maintain accurate and up-to-date parking information.

Actuators/Effectors:

Actuators are represented through the actions performed by the system after analyzing parking availability data. When a parking space status changes, the system updates the parking database to maintain real-time accuracy and ensure consistency across all system components. The application then displays the current availability of parking spaces to users through a visual interface, allowing drivers to easily identify which parking slots are occupied or available. When a driver requests parking assistance, the system generates navigation guidance by identifying the nearest available parking space based on the driver's location and providing directional instructions to guide the user efficiently to the selected slot. Additionally, if all parking spaces are occupied, the system triggers a notification to inform

users that the parking area is full, enabling them to avoid unnecessary entry and consider alternative parking locations. These actuators demonstrate how the intelligent agent responds to environmental inputs by updating information, guiding users to parking locations, and issuing alerts.

Sensors:

Sensors are represented as the input mechanisms that allow the intelligent agent to perceive its environment and receive user interactions. In the intelligent parking system, the sensors include surveillance cameras, parking occupancy sensors, driver location input (GPS), user interaction inputs from the application interface and database status logs that record changes in parking space occupancy. Parking space occupancy is captured through surveillance cameras or parking occupancy sensors, where each parking slot's status is identified as either occupied or vacant. The system also receives the driver's location information, which is used to determine the nearest available parking space. In addition, user actions, such as requesting parking availability or refreshing the system, serve as additional sensor inputs that trigger system responses. All detected changes and user interactions are recorded in the database to support continuous monitoring of the parking environment. These sensors enable the intelligent agent to gather relevant data, which forms the basis for intelligent reasoning, decision-making, and subsequent system actions.

4.0 Behaviours Of Agent To Achieve The System Goals

The proposed solution of the AI Smart Parking Detection System as an intelligent agent that is able to continuously perceive its environment to make decisions based on real time data and perform necessary actions to assist drivers in locating parking space more efficiently. The main goal of the agent is to provide accurate and timely information on parking availability which can vastly help in reducing parking search time and improve overall user experience in urban or campus environments.

For starters, the intelligent agent begins its operation by perceiving the environment through multiple sensors, including surveillance cameras, parking occupancy sensors, GPS location data from drivers and user interaction inputs from the mobile application. These sensors allow the agent to monitor and deduce parking space conditions, every tracked vehicle movement and user requests in real time. The agent continuously collects and updates this data to maintain precise representations of the current parking environment.

Once the data is received, the agent applies knowledge representations and state space reasoning to interpret the current status of each parking space. For instance, when no vehicle is detected in a parking slot, the agent temporarily marks the space as available and starts a time-based validation process to confirm its availability. This reasoning process helps the agent avoid false detections caused by environmental noise such as shadows or brief obstructions.

Based on the interpreted information, the agent then makes decisions and triggers appropriate actions through its actuators. When an available parking space is confirmed, the agent updates the parking database and displays the latest availability information on the application interface. If a driver requests parking assistance, the agent evaluates the driver's location and identifies the nearest available parking space, providing navigation guidance to direct the driver efficiently to the selected slot.

In situations where all parking spaces are occupied, the agent demonstrates adaptive behavior by sending timely notifications to inform drivers that the parking area is full. This prevents unnecessary entry into congested parking zones and allows drivers to make alternative decisions, such as searching for another parking location. As vehicles enter or leave parking spaces, the agent continuously updates the system state to reflect the latest current conditions.

The intelligent agent operates in a continuous perception–decision–action loop, allowing it to adapt dynamically to changes in the parking environment. By constantly monitoring sensor inputs to reasoning about parking availability in order to execute appropriate actions, the agent ensures that drivers receive reliable and up-to-date parking information. These behaviors collectively enable the system to achieve its goals of efficient parking detection which reduces search time, and improves urban parking management.